2014-2015 – Toulouse School of Economics – Master 2 Macro I Syllabus

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Objective of the course is

- to present the main theories of macroeconomics in the long run, i.e. the theory of economic growth and unemployment,
- to introduce the main facts of the business cycles, and the skeleton of dynamic stochastic general equilibrium models.

In our way, we'll introduce/review a number of tools in dynamic optimization, equilibrium derivation, and welfare analysis.

Prerequisites are : Kuhn-Tucker optimization theory ; standard calculus and linear algebra ; standard micro price theory.

Grading Policy : 100% for a 3 hours final exam

Useful textbooks are:

- Acemoglu, Introduction to modern economic growth, MIT press
- Ljungqvist and Sargent, Recursive Macroeconomic Theory , MIT Press
- Pissarides, Equilibrium Unemployment Theory

And also

- Blanchard and Fischer, Lectures on macroeconomics, MIT Press
- Romer, Advanced Macroeconomics, McGraw Hill
- Aghion and Howitt, The Economics of Growth, MIT Press
- Layard-Nickell-Jackman, Unemployment, Oxford

Course outline (slides of the course can be found at http://fportier.wordpress.com/teachingenseignement/m2-macro-course/)

- Lecture 1 : Optimal Control
- Lecture 2 : Balance Growth Paths
- Lecture 3: The Ramsey Growth Model
- Lecture 4: The Dixit-Stiglitz preference structure
- Lecture 5: Sustained Growth
- Lecture 6: R&D based models of endogenous growth
- Lecture 7: Computing the speed of convergence
- Lecture 8: Empirical studies of convergence
- Lecture 9: Asset values
- Lecture 10: The Mortensen-Pissarides matching model
- Lecture 11: The Hosios efficiency condition
- Lecture 12: Competitive search equilibrium
- Lecture 13: Solving Dynamic Rational Expectations Models
- Lecture 14: Business Cycle Facts
- Lecture 15: Real Business Cycle Models

Useful references to read are:

1. The Ramsey growth model
   - Tool : optimal control
   - Readings :
     - Blanchard/Fischer, chapter 2
     - Acemoglu, chapter 7
     - Acemoglu, chapter 8
     - Ramsey (1928), " A mathematical theory of saving ", Economic Journal
2. Innovation and endogenous growth
   - Tool : the Dixit-Stiglitz aggregative model
   - Readings :
     - Acemoglu, chapters 12 and 13
     - Dixit-Stiglitz, " Monopolistic Competition and Optimum Product Diversity" The American Economic Review, Vol. 67, No. 3 (Jun., 1977), pp. 297-308
     - Romer, Paul, ".Endogenous Technological Change ", Journal of Political Economy, Vol. 98, No. 5 (Oct., 1990), pp. S71-S102

  - Grossman-Helpman, " Quality Ladders in the Theory of Growth" Review of Economic Studies, Vol. 58, No. 1 (Jan., 1991), pp. 43-61
  - Aghion-Howitt, " A Model of Growth Through Creative Destruction" Econometrica, Vol. 60, No. 2 (Mar., 1992), pp. 323-351

3. Empirical growth

   - Tools : Log-linearization around steady state; computing a speed of convergence
   - Readings :
     - Acemoglu, chapter 3, 4
     - Mankiw, Romer, and Weil (1992), " A contribution to the empirics of economic growth ", Quarterly Journal of Economics
     - Barro and Sala-i-Martin (1991), " Convergence ", Journal of Political Economy
     - Jones (1995) " Time Series Tests of Endogenous Growth Models" Quarterly Journal of Economics, Vol. 110, No. 2 (May, 1995), pp. 495-525
     - Acemoglu, Johnson and Robinson (2001), " The Colonial Origins of Comparative Development: An Empirical Investigation" The American Economic Review, Vol. 91, No. 5 (Dec., 2001), pp. 1369-1401

4. The Mortensen/Pissarides matching model

   - Tool : Comparing optimum and equilibrium in dynamic models with market failures
   - Readings :
     - Pissarides, C. (1990) Equilibrium Unemployment Theory, 2nd edition, MIT Press, 1998
     - Mortensen, D. and C. Pissarides, (1994) "Job Creation and Job Destruction in the Theory of Unemployment", Review of Economic Studies; 61(3), pages 397-415.
     - Hosios, Arthur (1990), " On the Efficiency of Matching and Related Models of Search and Unemployment" Review of Economic Studies, Vol. 57, No. 2 (Apr., 1990), pp. 279-298
     - Moen, Ragnar (1995), " Competitive search equilibrium ", Journal of Political Economy

5. Business Cycle facts

   - Tool : Spectral Analysis, National Accounting
   - Readings :
     - Cochrane, John, "Time Series for Macroeconomics and Finance", lectures notes
     - Granger, C, "The Typical Spectral Shape of an Economic Variable", Econometrica, 1966
     - Baxter, Marianne and Robert G. King. "Measuring Business Cycles: Approximate Band-Pass Filters For Economic Time Series," Review of Economics and Statistics, 1999,
     - Christiano and Fitzgerald, 2003. "The Band Pass Filter," International Economic Review
     - Canova, Fabio, 1998. "Detrending and business cycle facts," Journal of Monetary Economics

  - Burnside, 1998. "Detrending and business cycle facts: A Comment," Journal of Monetary Economics
  - Ravn, Morten O. and Uhlig, Harald, 2001. "On Adjusting the HP-Filter for the Frequency of Observations," CEPR Discussion Papers 2858,
  - NIPA Handbook: Concepts and Methods of the U.S. National Income and Product Accounts, 2011, Bureau of Economic Analysis website

6. Real Business Cycles

- Tool : Linearization, General Equilibrium
- Readings :
  - Kydland, Finn E and Prescott, Edward C, 1982. "Time to Build and Aggregate Fluctuations," Econometrica
  - King, Robert G. and Plosser, Charles I. and Rebelo, Sergio T., 1988. "Production, growth and business cycles : I. The basic neoclassical model," Journal of Monetary Economics,
  - King, Robert G. and Plosser, Charles I. and Rebelo, Sergio T., 1988. "Production, growth and business cycles : II. New directions," Journal of Monetary Economics,
  - King, Robert G. and Rebelo, Sergio T., 1999. "Resuscitating real business cycles," Handbook of Macroeconomics, in: J. B. Taylor and M. Woodford (ed.), Handbook of Macroeconomics, edition 1, volume 1, chapter 14, pages 927-1007 Elsevier.
  - Rogerson, Richard, 1988. "Indivisible labor, lotteries and equilibrium," Journal of Monetary Economics
  - Hansen, Gary D., 1985. "Indivisible labor and the business cycle," Journal of Monetary Economics